Hexapod Robot

BY GONE RAMU

Project Aim

- To control a Hexapod Robot by mobile using HC – 05 Bluetooth Module
- To Familarize with Scratch software by MIT

Apparatus Required

Hardware Tools :

1. Arduino Mega
2. Bluetooth HC – 05 module
3. 3 Servo Motors
4. Jumper wires
5. Arduino & USB cable
6. 2 Battery Source
7. Screws and nuts
8. Clamps

Apparatus Required

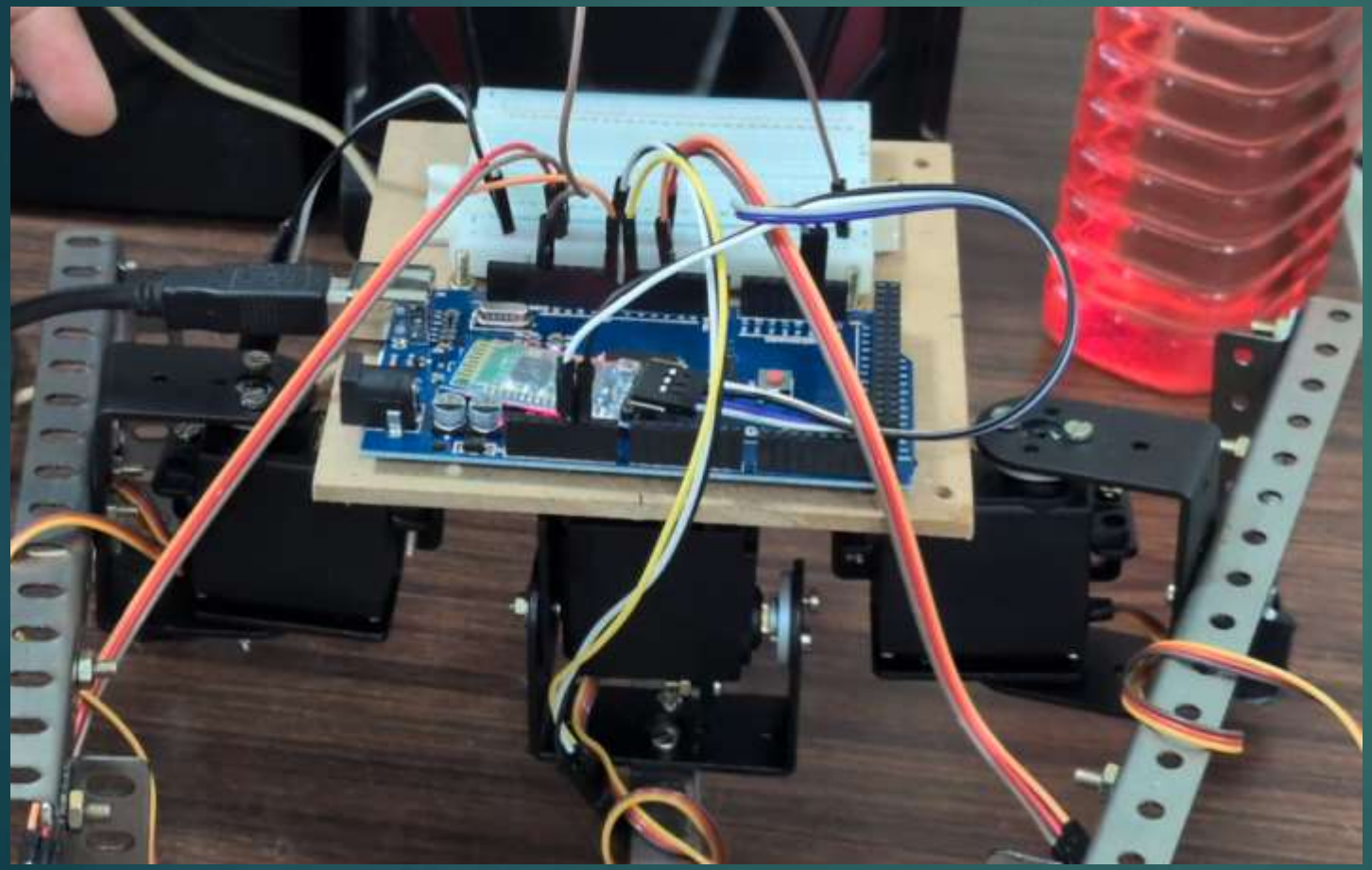
Software Tools :

1. Arduino IDE
2. UI Interface
3. Scartch Software by MIT
4. Arduino code of Hexapod Robot
5. Arduino code of Bluetooth Hexapod Robot
6. Scratch Software code
7. Robotic Arm cad file

Procedure

- Arrange 3 Servos in according to Hexapod Robot structure along with its parts.
- Integrate 4 Servos with Arduino mega.
- Now connect the Bluetooth HC – 05 module to Arduino mega.
- Connect Batteries to Servos for supplying the power.
- Open the Arduino IDE
- Upload the Hexpod Robot Code
- Turn on the Bluetooth in your mobile and open UI application.
- Click on connect. Click on “HC – 05 Bluetooth module”
- Bluetooth connection is Done.
- Similarily, you can run robot using Scratch code (mit software) which uses blocks instead of code.

Circuit Connection



Servos connection with Batteries :

4 Servos	Battery
VCC	Anode
GND	Cathode/GND

Servos connection with Arduino mega :

Servo signal pin	Arduino mega
Front servo	4
Lift Servo	7
Rear Servo	8

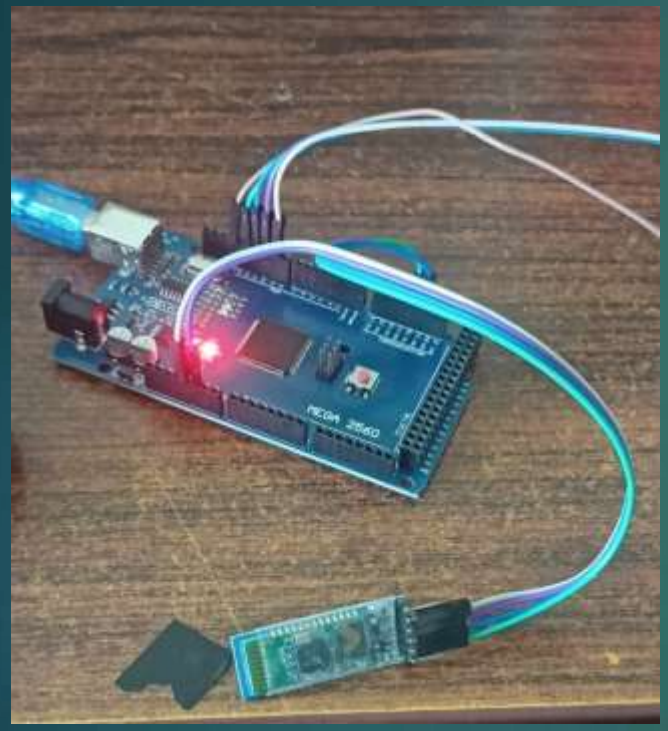
Circuit Connection



Integration of Bluetooth module with Arduino mega :

Bluetooth module	Arduino mega
TX	RX (19)
RX	TX (18)
VCC	5v
GND	GND

UI Application Connection



Bluetooth connection



UI interface by Skyfi labs

Hexapod Walking Gait

Forward Motion

Step	Which Servo/ Leg to actuate?	Angle which Servo Arm should move to
Lift Left Side of Body Up	Middle	120 degrees
Pull Forward using Right front and Right Rear Legs which are in contact with ground	Front and Rear	60 degrees
Bring all legs Down – Bring body to level position	Middle	90 degrees
Lift Right Side of the Body Up	Middle	60 degrees
Pull using the Left Front and Left Rear legs which are on the ground – this brings them to neutral position	Front and Rear	90 degrees
Bring all legs Down – Body to level position	Middle	90 degrees

Backward Motion

Step	Which Servo/ Leg to actuate?	Angle which Servo Arm should move to
Lift Left Side of Body Up	Middle	120 degrees
Push Backwad using Right front and Right Rear Legs which are in contact with ground	Front and Rear	120 degrees
Bring all legs Down – Bring body to level position	Middle	90 degrees
Lift Right Side of the Body Up	Middle	60 degrees
Push using the Left Front and Left Rear legs which are on the ground – this brings them to neutral position	Front and Rear	90 degrees
Bring all legs Down – Body to level position	Middle	90 degrees

Result

- Finally, we made Hexapod Robot

using Arduino Mega & Bluetooth module

